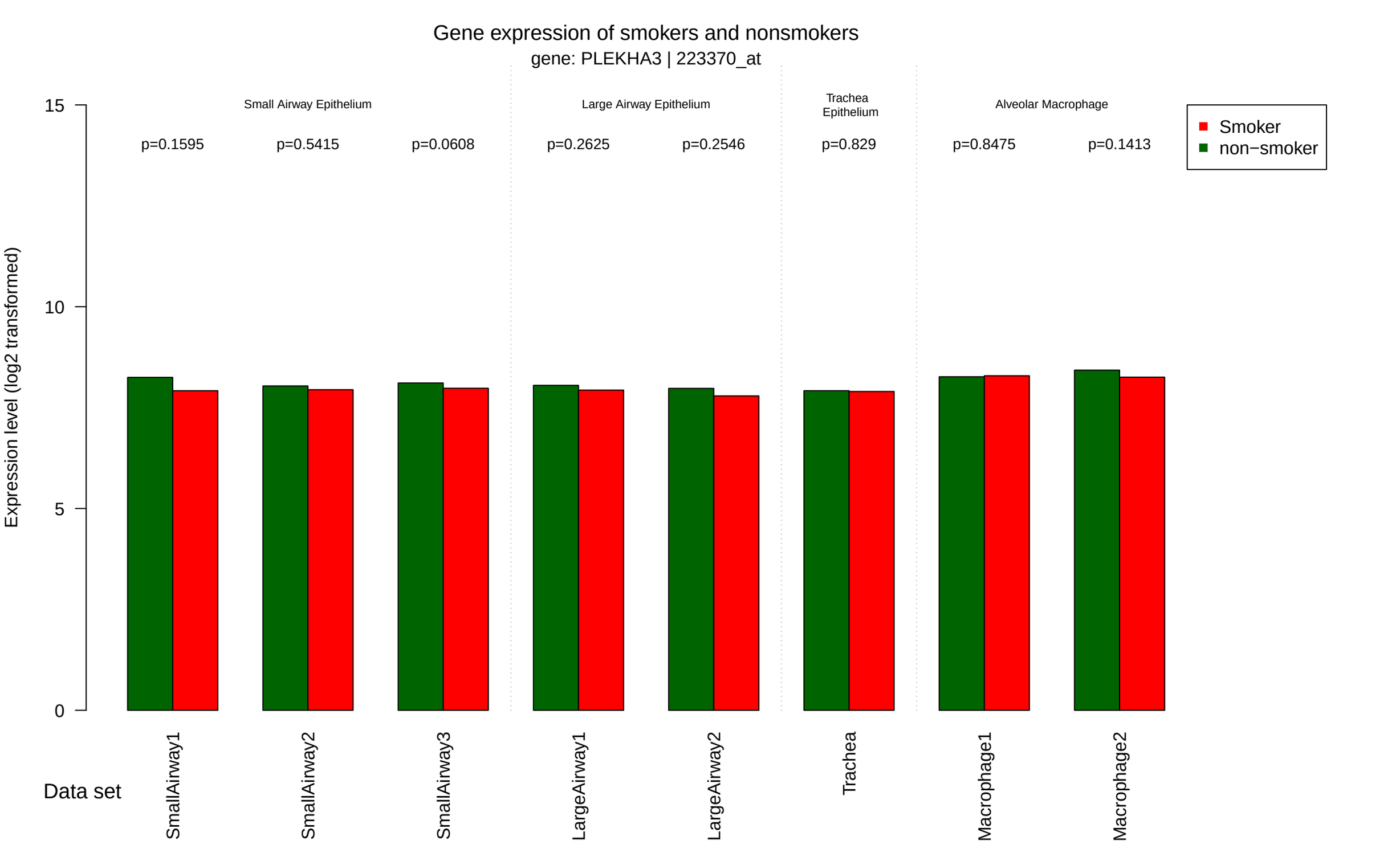
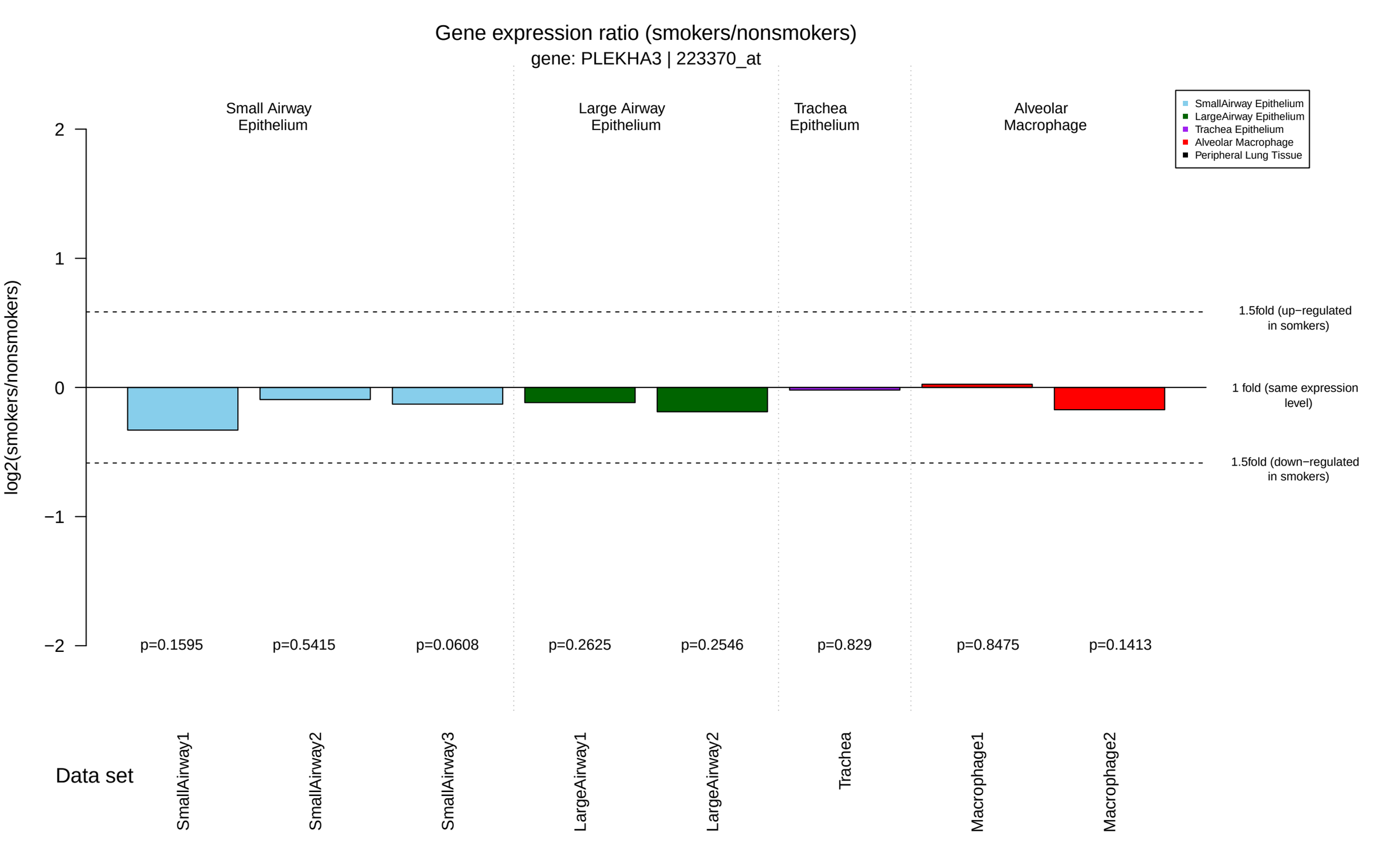
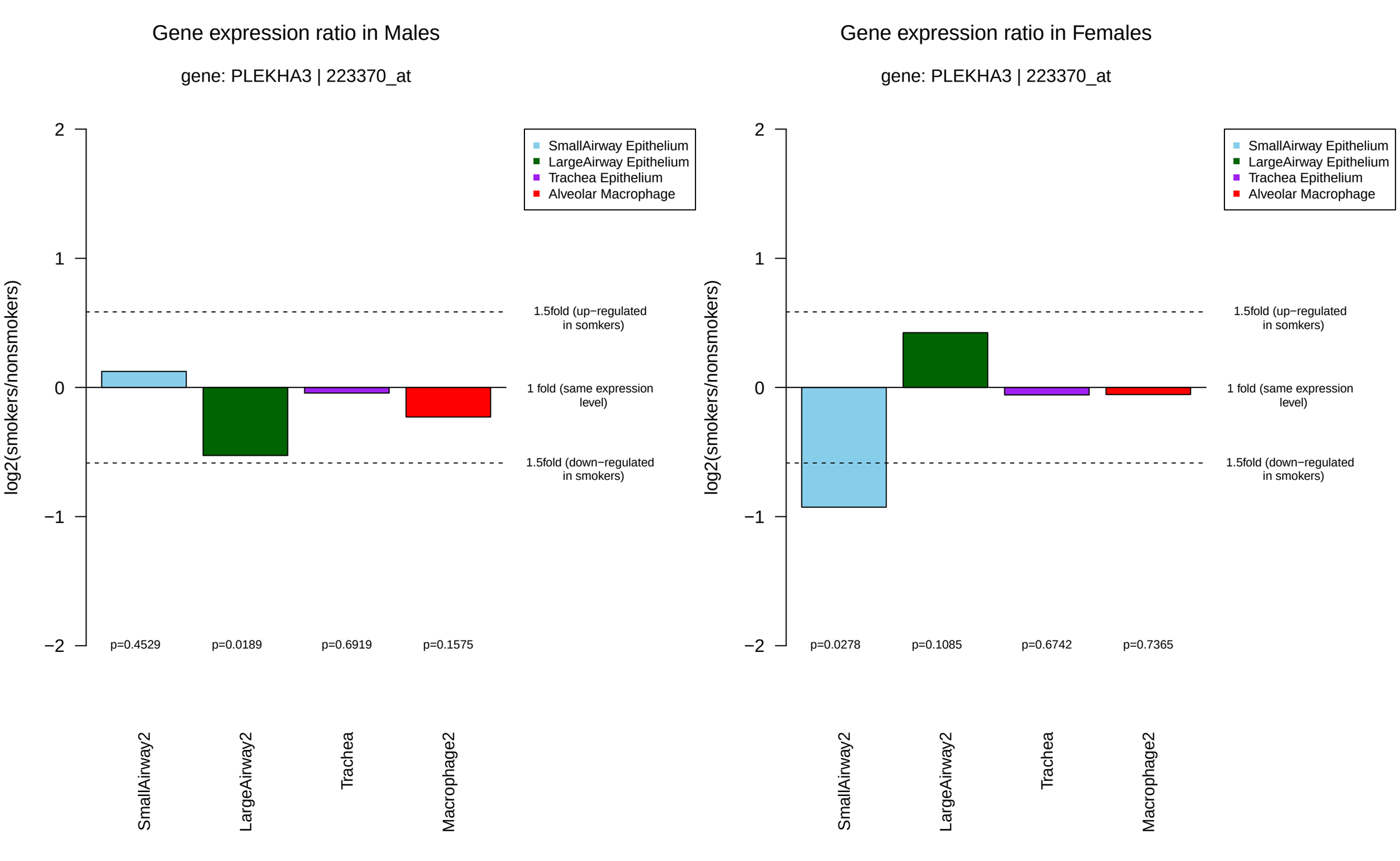
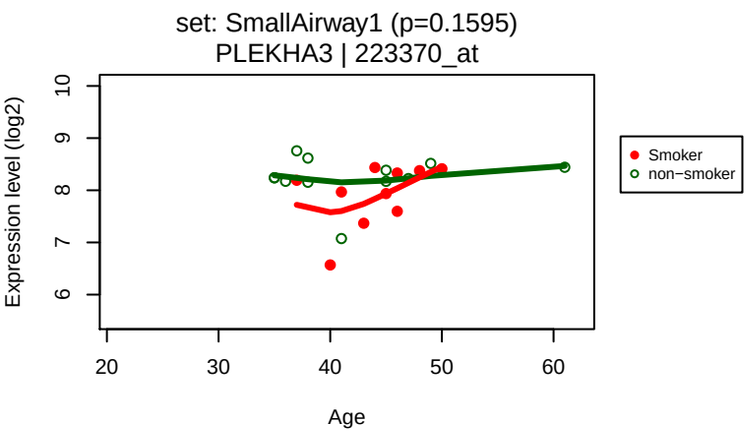
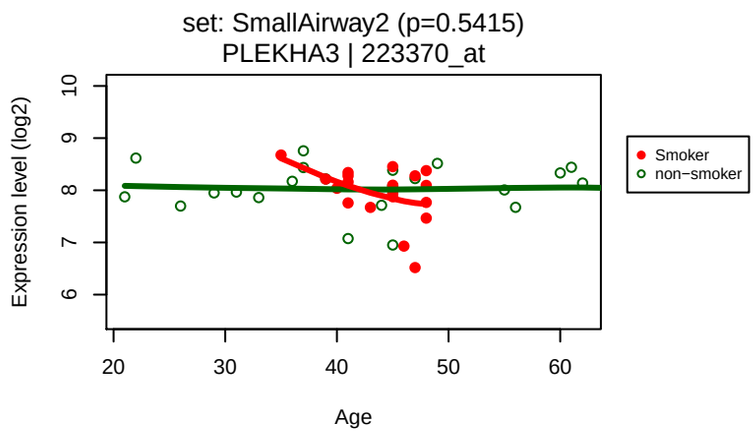
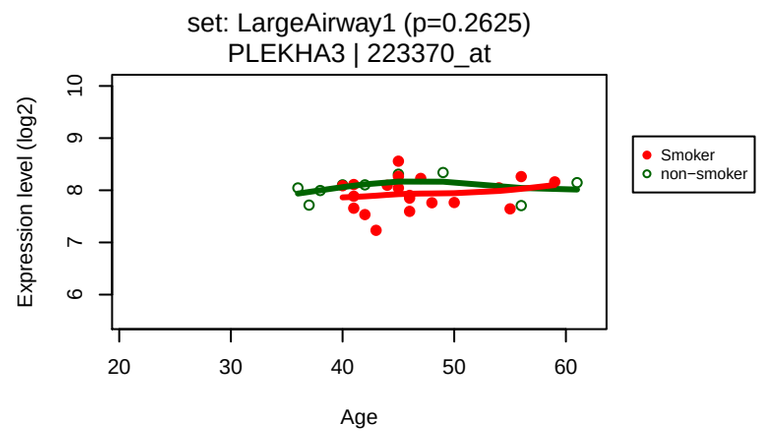

Gene expression of smokers and nonsmokers
gene: PLEKHA3 | 223370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1595
p=0.5415
p=0.0608
p=0.2625
p=0.2546
p=0.829
p=0.8475
p=0.1413
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

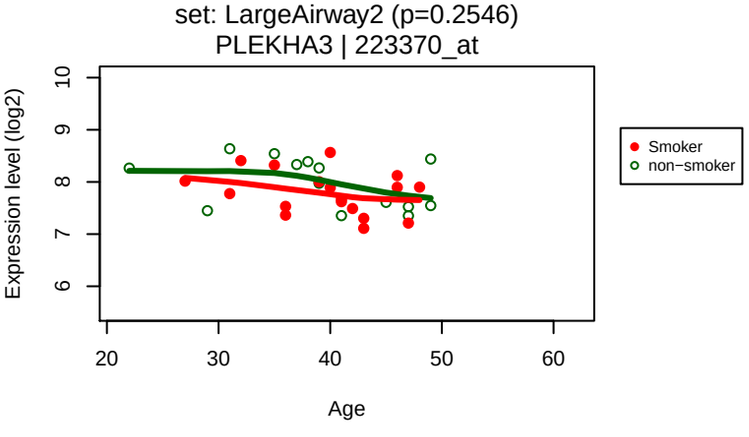

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA3 | 223370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1595
p=0.5415
p=0.0608
p=0.2625
p=0.2546
p=0.829
p=0.8475
p=0.1413
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

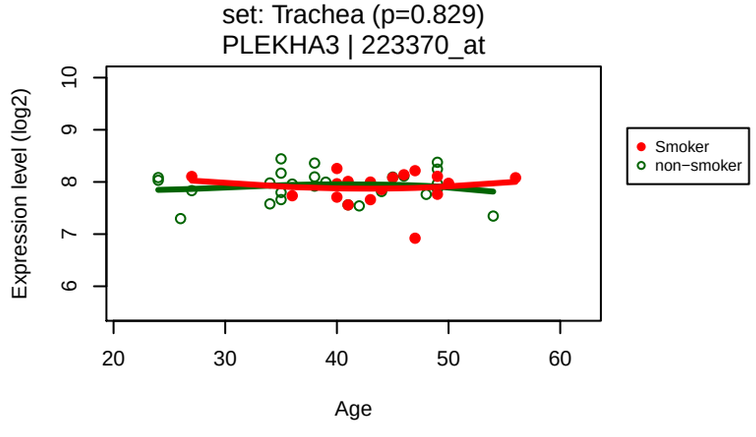

Gene expression ratio in Males
gene: PLEKHA3 | 223370_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4529
p=0.0189
p=0.6919
p=0.1575
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA3 | 223370_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0278
p=0.1085
p=0.6742
p=0.7365
SmallAirway2
LargeAirway2
Trachea
Macrophage2

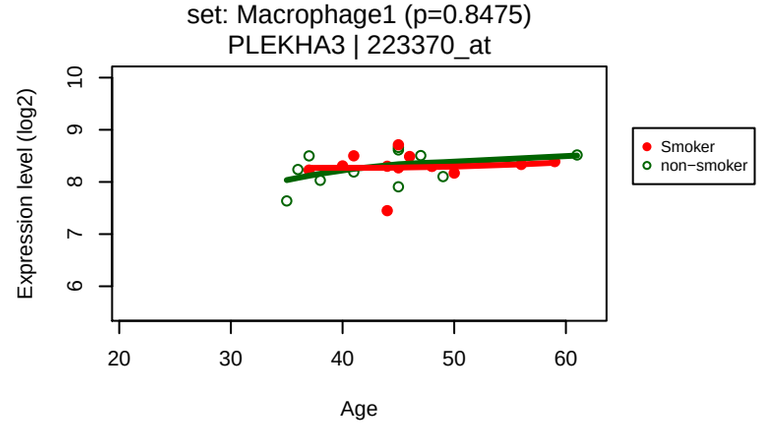

set: SmallAirway1 (p=0.1595)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker

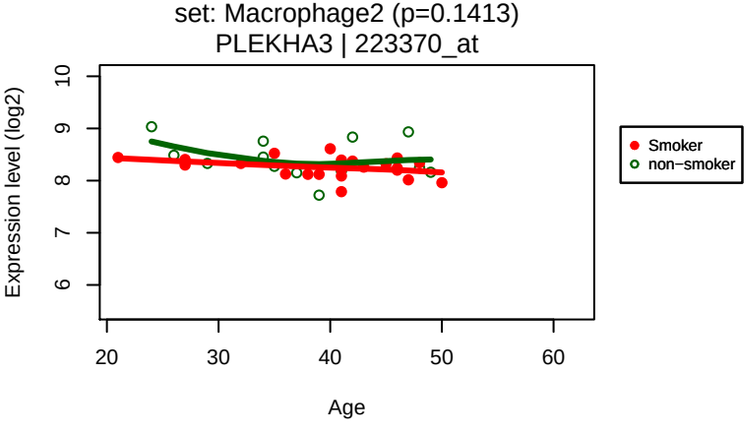

set: SmallAirway2 (p=0.5415)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker

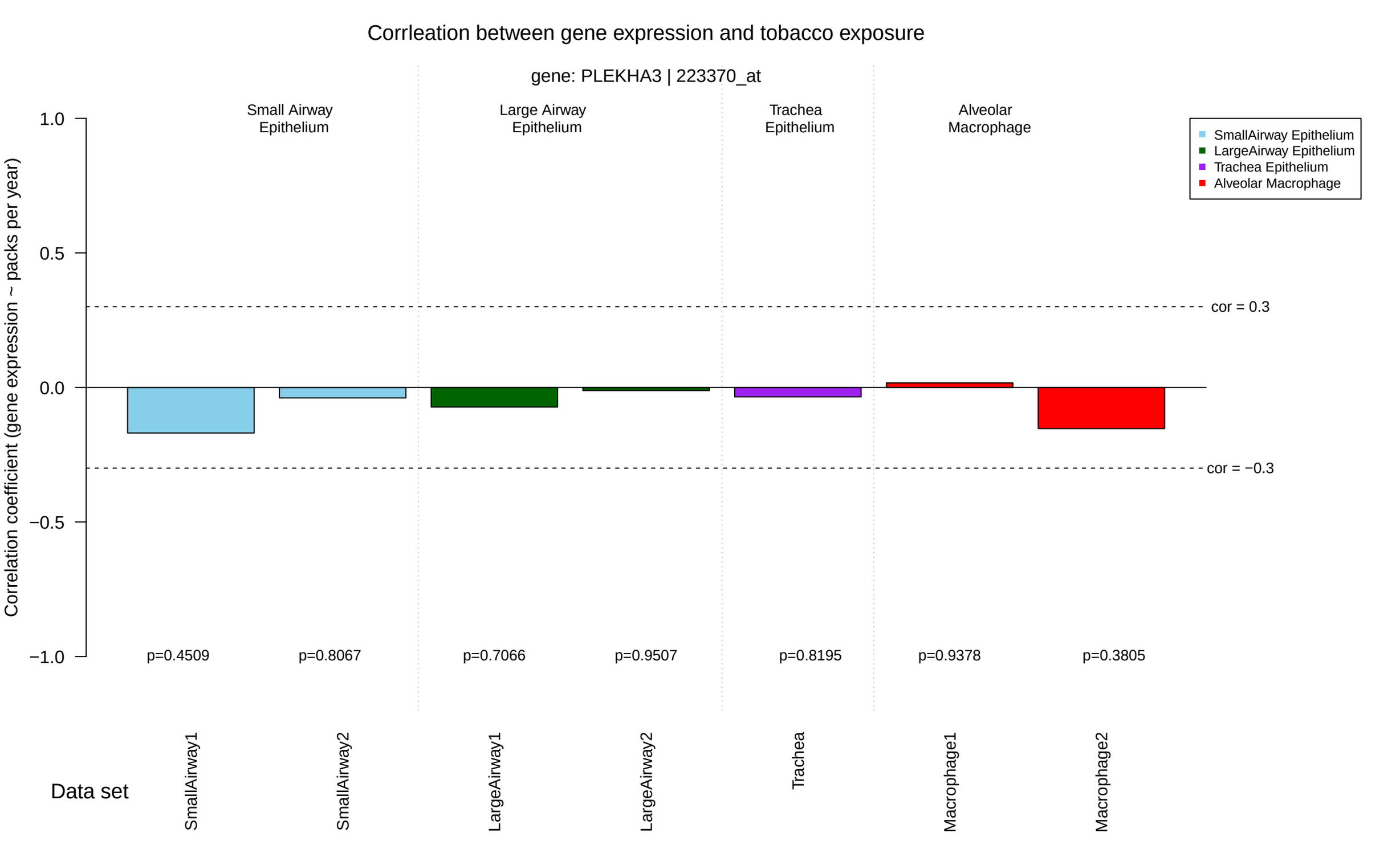

set: LargeAirway1 (p=0.2625)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2546)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.829)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8475)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1413)
PLEKHA3 | 223370_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA3 | 223370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4509
p=0.8067
p=0.7066
p=0.9507
p=0.8195
p=0.9378
p=0.3805
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2